

Project Title: Data Analyst – Securitisation

Date:22-06-2026

Executive Summary ;

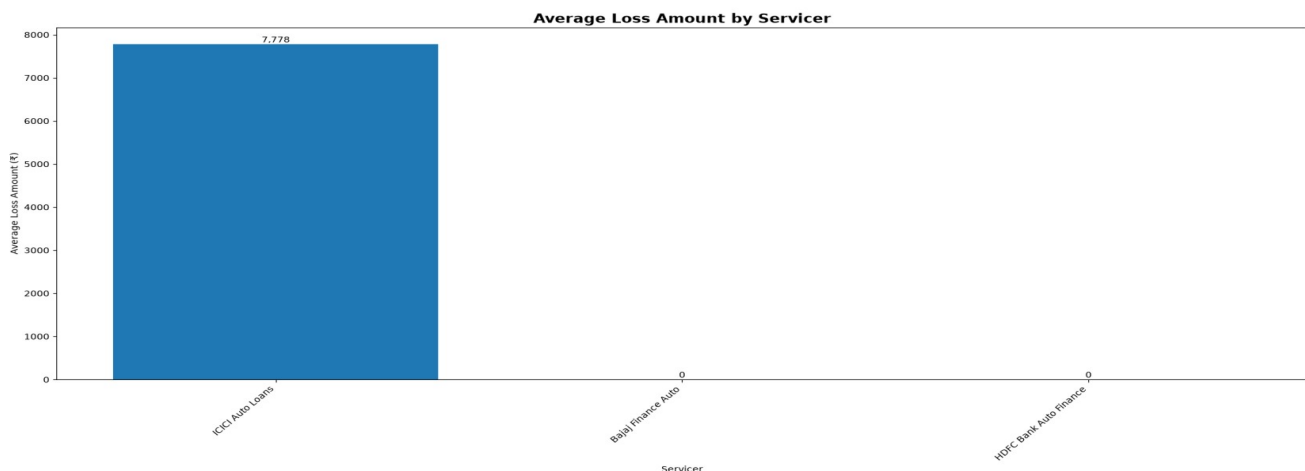
"The primary objective of this project is to perform a comprehensive risk assessment of the securitisation portfolio. This project involves data loading, the development of risk metrics using advanced DAX measures, and the preparation of professional reports for investors."

Day 1: Data Analysis Findings:

"On the first day of the project, I successfully loaded the primary dataset (`data.csv`) into the analysis environment using Python's Pandas library to initiate the securitisation data exploration. The key findings from the initial analysis are as follows:

- **Portfolio Distribution:** The dataset comprises loan records from three major servicers: Bajaj Finance Auto, ICICI Auto Loans, and HDFC Bank Auto Finance.
- **Key Insight (Risk Identification):** The analysis revealed a significant variance in portfolio performance. Specifically, **ICICI Auto Loans** exhibits a higher average `LOSSAmount` compared to other servicers, whereas Bajaj Finance Auto and HDFC Bank Auto Finance currently show negligible loss figures.
- **Analytical Methodology:** I utilized the `groupby()` function to aggregate the data by `ServicerName` and calculated the mean `LOSSAmount` to identify these performance disparities. A bar chart was generated to visually represent these findings, providing a clear indication of potential credit risk concentration within the ICICI Auto Loans segment."

Figure 1



Zetheta Project 1A – Loan Portfolio Risk Analysis Report

Cover Page

Project Name:

Loan Portfolio Risk Analysis

Submitted By:

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Tools Used:

Python, Pandas, Matplotlib, VS Code

Day 1 – Dataset Understanding & Data Loading

Objective

Understand the dataset structure and load the CSV file into Python.

Activities Performed

- Loaded dataset using Pandas
- Checked column names
- Explored dataset structure
- Verified data quality

Code Used

```
df = pd.read_csv("Data/data.csv")  
print(df.columns.tolist())
```

Screenshot 1

The screenshot shows a Jupyter Notebook titled 'Zetheta Project 1A - Loan Portfolio Risk Analysis'. The script defines a risk scoring model and analyzes loan data. The output includes a risk analysis report and a fact table.

```
1 # Zetheta Project 1A - Loan Portfolio Risk Analysis
2 ## Risk Scoring Model
3 ## Python Logic
4
5 def risk_analysis(loan_data):
6     # Risk Scoring Logic
7     # ...
8     return 'Low Risk'
9
10
11 ## Risk Analysis Results
12
13 # Example Output:
14 # Servicer-wise loan risk distribution was calculated.
15
16 # Example Output:
17
18 # Dimension Table (Servicer):
19 # ServicerID ServicerName
20 # 0 SVC001 HDFC Bank Auto Finance
21 # 1 SVC003 Bajaj Finance Auto
22 # 2 SVC002 ICICI Auto Loans
23
24 # Top 5 Servicers by Loss:
25 # ServicerName LossAmount
26 # ICICI Auto Loans 7778.21939
27 # Bajaj Finance Auto 0.00000
28 # HDFC Bank Auto Finance 0.00000
29 # Name: LossAmount, dtype: float64
30
31 --- Risk Analysis Report ---
32 Risk_Catagory Low Risk Medium Risk
33 ServicerName
34 Bajaj Finance Auto 145 35
35 HDFC Bank Auto Finance 137 19
36 ICICI Auto Loans 133 31
37
38 PS C:\Users\Vishnu\Desktop\Zetheta Project 1A> python Scripts\Python\analysis.py
39 [LoanID, 'PoolID', 'BorrowerID', 'ServicerID', 'ServicerName', 'OriginationChannel', 'OriginationDate', 'CutoffDate', 'MaturityDate', 'OriginalLoanAmount', 'OriginalTerm', 'RemainingTerm', 'MonthsOnBook', 'InterestRate', 'MonthlyEMI', 'LoanPurpose', 'CurrentBalance', 'ScheduledPaymentDue', 'LastPaymentAmount', 'LastPaymentDate', 'DelinquencyStatus', 'DelinquencyDays', 'TotalPaymentsDue', 'TotalPaymentsMade', 'Times30DPD_Last12M', 'Times60DPD_Last12M', 'Times90DPD_Last12M', 'VehicleMake', 'VehicleYear', 'VehicleType', 'IsNewVehicle', 'OriginalVehicleValue', 'CurrentVehicleValue', 'LTV_AtOrigination', 'LTV_Current', 'BorrowerAge', 'EmploymentType', 'AnnualIncome_Ink', 'DTI_Ratio', 'Credit_Score_Origination', 'Credit_Score_Current', 'Region', 'State', 'IPRS9_Stage', 'PO_Estimate', 'LGD_Estimate', 'EAD', 'Ed_Provision', 'IsDefaulted', 'IsModified', 'ModificationType', 'LossAmount', 'RecoveryAmount', 'NetLoss', 'PrepaymentAmount', 'InsuranceType', 'HasInsurance']
40 CSV Successfully Loaded
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42 Fact Table:
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44 LoanID PoolID BorrowerID ServicerID ServicerName OriginationChannel OriginationDate ... ModificationType LossAmount RecoveryAmount NetLoss PrepaymentAmount InsuranceType HasInsurance
45 0 ZAUTO02024-1-00001 ZAUTO02024-1 BOR-748564 SVC001 HDFC Bank Auto Finance Direct 2024-08-02 ... NaN 0.0 0.0 0.0 0.0 Comprehensive True
46 1 ZAUTO02024-1-00002 ZAUTO02024-1 BOR-748564 SVC003 Bajaj Finance Auto Online 2024-05-21 ... NaN 0.0 0.0 0.0 0.0 Comprehensive True
47 2 ZAUTO02024-1-00003 ZAUTO02024-1 BOR-694731 SVC002 ICICI Auto Loans Online 2024-08-03 ... NaN 0.0 0.0 0.0 0.0 Comprehensive True
48 3 ZAUTO02024-1-00004 ZAUTO02024-1 BOR-855731 SVC003 Bajaj Finance Auto Online 2023-05-15 ... NaN 0.0 0.0 0.0 0.0 Nil True
49 4 ZAUTO02024-1-00005 ZAUTO02024-1 BOR-857549 SVC003 Bajaj Finance Auto Online 2021-08-18 ... NaN 0.0 0.0 0.0 0.0 Nil True
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Day 2 – Loss Analysis

Objective

Analyze average loss amount by servicer.

Activities Performed

- Replaced missing LossAmount values
- Grouped data by ServicerName
- Calculated average loss
- Created bar chart

Screenshot 2

```
1 # Zetheta Project 1A - Loan Portfolio Risk Analysis
2 ## Risk Scoring Model
3 ## Python Logic
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5 if __name__ == '__main__':
6     return 'Low Risk'
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Observation

- Different servicers show different loss patterns.
- Some servicers have higher average losses.

Day 3 – Star Schema Design

Objective

Create analytical data model.

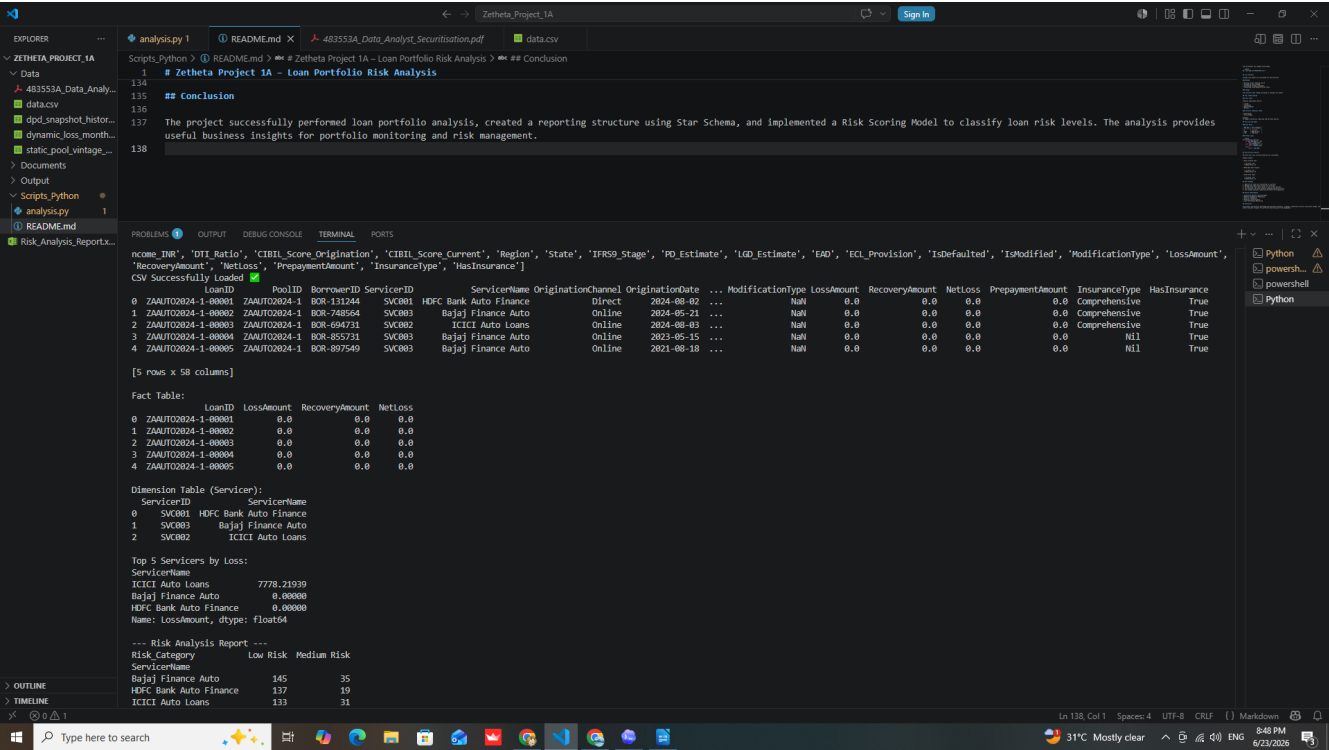
Fact Table

- LoanID
- LossAmount
- RecoveryAmount
- NetLoss

Dimension Table

- ServicerID
- ServicerName

Screenshot 3



Observation

Star Schema improves reporting and analytics performance.

Day 4 – Risk Scoring Model

Objective

Categorize loans based on DPD Days.

Risk Rules

DPD Days Risk Category

0 Low Risk

DPD Days Risk Category

1-30 Medium Risk

Above 30 High Risk

Screenshot 4

```
analysis.py 1 | README.md x | 483553A_Data_Analyt_Securitisation.pdf | data.csv
ZETHETA_PROJECT_1A
Data
483553A_Data_Analy...
data.csv
dpd-snapshot_histor...
dynamic_loss_month...
static_pool_vintage...
Documents
Output
Scripts_Python
analysis.py 1
README.md
Risk_Analysis_Report...

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

LoanID PoolID BorrowerID ServicerID ServicerName OriginationChannel OriginationDate ... ModificationType LossAmount RecoveryAmount NetLoss PrepaymentAmount InsuranceType HasInsurance
0 ZAAUTO2024-1-00001 ZAAUTO2024-1 BOR-131244 SVC001 HDFC Bank Auto Finance Direct 2024-08-02 ... NaN 0.0 0.0 0.0 0.0 Comprehensive True
1 ZAAUTO2024-1-00002 ZAAUTO2024-1 BOR-748564 SVC003 Bajaj Finance Auto Online 2024-05-21 ... NaN 0.0 0.0 0.0 0.0 Comprehensive True
2 ZAAUTO2024-1-00003 ZAAUTO2024-1 BOR-694731 SVC002 ICICI Auto Loans Online 2024-08-03 ... NaN 0.0 0.0 0.0 0.0 Comprehensive True
3 ZAAUTO2024-1-00004 ZAAUTO2024-1 BOR-855731 SVC003 Bajaj Finance Auto Online 2023-05-15 ... NaN 0.0 0.0 0.0 0.0 N/A True
4 ZAAUTO2024-1-00005 ZAAUTO2024-1 BOR-897549 SVC003 Bajaj Finance Auto Online 2021-08-18 ... NaN 0.0 0.0 0.0 0.0 N/A True

[5 rows x 58 columns]

Fact Table:
LoanID LossAmount RecoveryAmount NetLoss
0 ZAAUTO2024-1-00001 0.0 0.0 0.0
1 ZAAUTO2024-1-00002 0.0 0.0 0.0
2 ZAAUTO2024-1-00003 0.0 0.0 0.0
3 ZAAUTO2024-1-00004 0.0 0.0 0.0
4 ZAAUTO2024-1-00005 0.0 0.0 0.0

Dimension Table (Servicer):
ServicerID ServicerName
0 SVC001 HDFC Bank Auto Finance
1 SVC003 Bajaj Finance Auto
2 SVC002 ICICI Auto Loans

Top 5 Servicers by Loss:
ServicerName
ICICI Auto Loans 7778.21939
Bajaj Finance Auto 0.00000
HDFC Bank Auto Finance 0.00000
Name: LossAmount, dtype: float64

--- Risk Analysis Report ---
Risk Category Low Risk Medium Risk
ServicerName
Bajaj Finance Auto 145 35
HDFC Bank Auto Finance 137 19
ICICI Auto Loans 133 31
```

Observation

- Majority loans are Low Risk.
- No High Risk loans identified.

Day 5 – Business Insights

Key Findings

- Majority loans belong to Low Risk category.
- Bajaj Finance Auto has highest Medium Risk loans.
- HDFC Bank Auto Finance has lowest Medium Risk loans.
- No High Risk loans found.

5. Portfolio appears stable.

Business Recommendations

1. Continue monitoring Medium Risk loans regularly.
 2. Improve collection efforts for overdue accounts.
 3. Develop predictive models for early default detection.
 4. Create dashboards for real-time portfolio monitoring.
 5. Use risk-based segmentation for better decision-making.
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References

- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Loan Securitization Dataset (Provided by Zetheta)

Conclusion

The project successfully analyzed the loan portfolio dataset using Python and Pandas. Loss Analysis, Star Schema Design, and Risk Scoring Model were implemented successfully. The portfolio shows a healthy distribution of loans with low overall risk.

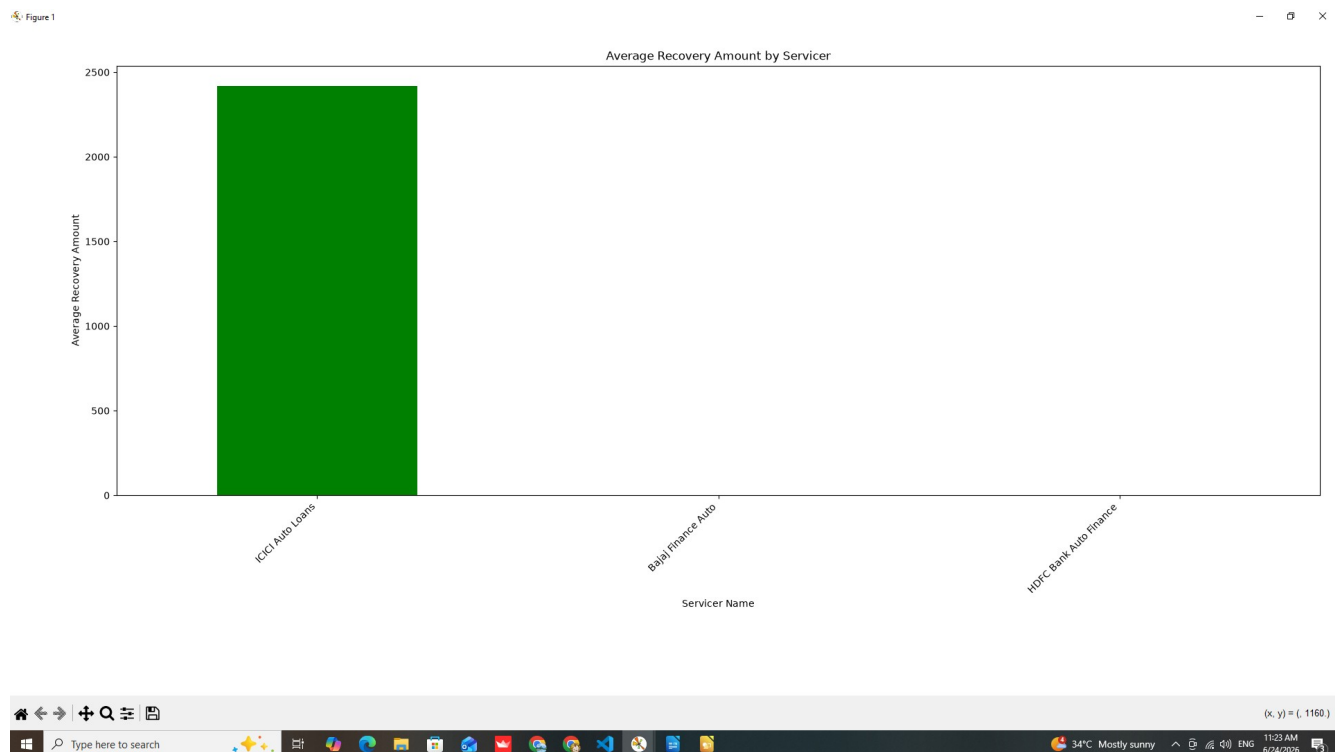
Future Improvements

- Machine Learning Model
- Default Prediction
- Power BI Dashboard
- Recovery Analysis
- Loan Monitoring System

Day 6 – Recovery Analysis

Recovery Amount analysis was performed to identify which servicer recovered the highest amount from defaulted loans.

The average recovery amount was calculated using Pandas groupby() function and visualized using a bar chart.

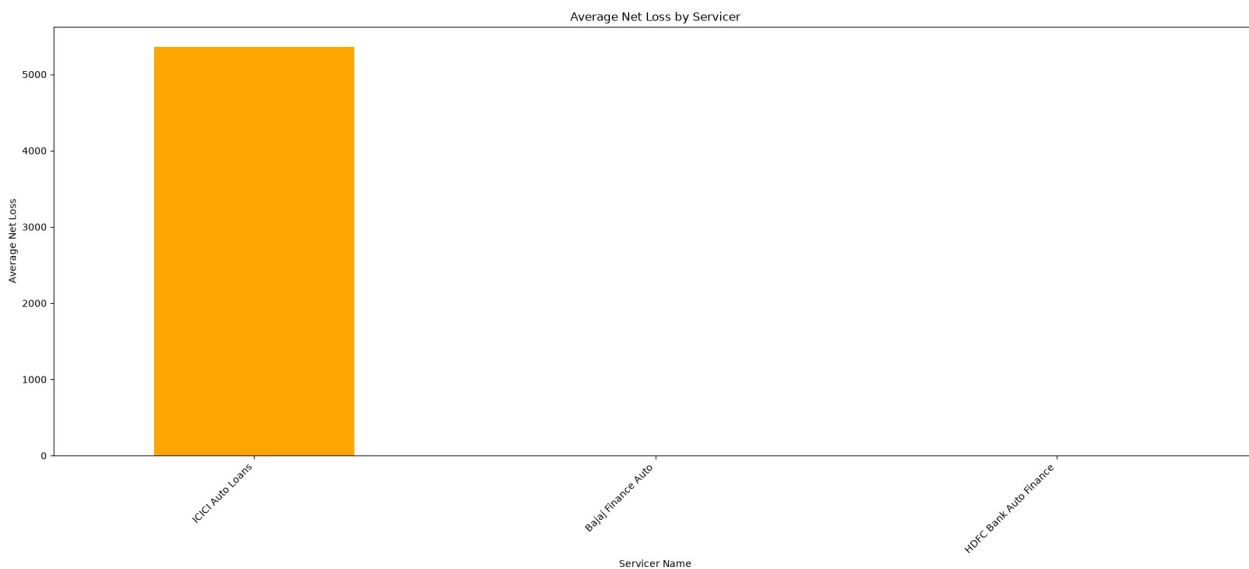


Observation

- ICICI Auto Loans shows the highest recovery amount.
- Bajaj Finance Auto and HDFC Bank Auto Finance show negligible recovery values.

Day 7 – Net Loss Analysis

Figure 1



Observation

ICICI Auto Loans has the highest average net loss.

Bajaj Finance Auto and HDFC Bank Auto Finance show negligible net loss.

Net loss analysis helps identify high-risk servicers.